

Fourier Series (Part 3)

Bessel inequality, Parseval condition

Least Square Approximation

for Electronics and Telecommunications

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March 28, 2026

- Orthogonal and orthonormal sequence and series (revision),
- The best mean square approximation - Euler-Fourier coefficients (revision),
- Problem of the least square approximation,
- Parseval's condition (ideal situation),
- Bessel inequality (realistic approximation),
- Orthonormal complete system.

Orthogonal Sequence of Functions

The sequence of functions:

$$(\varphi_n(x)) = \varphi_0(x), \varphi_1(x), \varphi_2(x), \dots$$

is called **orthogonal on the interval** $\langle a, b \rangle$ if:

- (1) $\forall m, n \in \mathbb{N}, m \neq n : \langle \varphi_m, \varphi_n \rangle = 0$ (any two are orthogonal)
- (2) $\forall : \|\varphi_n\|_2 > 0$ (no zero functions in this sequence)

Orthonormal sequence of functions

Sequence of the functions $(\varphi_n(x))$ is called **orthonormal** if:

$$\langle \varphi_n, \varphi_m \rangle = \delta_{nm} = \begin{cases} 0 & \text{if } n \neq m \\ 1 & \text{if } n = m \end{cases}$$

(δ_{nm} is called Kronecker delta).

In other words, the sequence of functions is orthogonal and the norm of each of those functions is 1, then it is orthonormal.

Orthogonal Series

Let $(\varphi_n(x))$ be an orthogonal sequence on $\langle a, b \rangle$ and (γ_n) be any number sequence then

$$\sum_{n=0}^{\infty} \gamma_n \varphi_n(x)$$

is an **orthogonal series***.

* It should be formally written as $\sum_{n=0}^{\infty} \gamma_n \varphi_n(\cdot)$, rather than $\sum_{n=0}^{\infty} \gamma_n \varphi_n(x)$ (but the latter notation is broadly used by engineers).

Convergence of an orthogonal series (revision)

Lemma (Best mean square approx is with Euler-Fourier coefficients)

If f is square integrable on $\langle a, b \rangle$ and the following orthogonal series is converging **in the mean square norm** on the interval $\langle a, b \rangle$ to f ie.

$$\int_a^b \left(\sum_{n=0}^N \gamma_n \varphi_n(x) - f(x) \right)^2 dx \xrightarrow{N \rightarrow +\infty} 0, \text{ then } \gamma_n = \frac{\langle f, \varphi_n \rangle}{\|\varphi_n\|_2^2}.$$

In a different notation same can be written as

$$d_2 \left(\sum_{n=0}^N \gamma_n \varphi_n, f \right) \xrightarrow{N \rightarrow +\infty} 0 \text{ then } \gamma_n = \frac{\langle f, \varphi_n \rangle}{\|\varphi_n\|_2^2}.$$

Can we take truncate Fourier series into a finite sum and still hope to get a good approximation?

This questions boils down to asking how fast is the mean square convergence. Good news for the computer calculations (which are finite and inaccurate by its nature) is that the speed of convergence is fast, and the mathematically rigorous Theorem About the Least Square Approximation justifies computational approach.

To state the problem more precisely let's define

Orthogonal Polynomial

$\delta_N(x) = \sum_{n=0}^N \gamma_n \varphi_n(x)$ is an **orthogonal polynomial** of degree N where γ_n - numbers and $(\varphi_n(x))$ - orthogonal sequence.

Given a square integrable function f on $\langle a, b \rangle$ and an orthogonal sequence $(\varphi_m(x))$ on $\langle a, b \rangle$ we are searching for coefficients γ_n of the orthogonal polynomial $\delta_N(x)$ of degree N such that the distance $\rho_2(\delta_N, f)$ is minimal.



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Why can we just take finite sums in a Fourier series and still get a good approximation?

Theorem About the Least Square Approximation

Among all polynomials of the form

$$\sum_{n=0}^N \gamma_n \varphi_n(x) = \delta_N(x)$$

the least square approximation of the function f over the interval $\langle a, b \rangle$ is the polynomial of the coefficients γ_n given by the Euler-Fourier formula.

We have to check when $\rho_2(\delta_N, f)$ is minimal. But it's easier to find the minimum of $\rho_2^2(\delta_N, f)$

$$\begin{aligned}
 \rho_2^2(\delta_N, f) &= \langle f - \delta_N, f - \delta_N \rangle = \langle f, f \rangle - 2 \langle f, \delta_N \rangle + \langle \delta_N, \delta_N \rangle = \\
 &= \langle f, f \rangle - 2 \left\langle f, \sum_{n=0}^N \gamma_n \varphi_n \right\rangle + \langle \delta_N, \delta_N \rangle = \\
 &= \langle f, f \rangle - 2 \sum_{n=0}^N \gamma_n \langle f, \varphi_n \rangle + \sum_{n=0}^N \gamma_n^2 \|\varphi_n\|_2^2 = \\
 &= \|f\|_2^2 - \sum_{n=0}^N 2\gamma_n c_n \|\varphi_n\|_2^2 + \sum_{n=0}^N \gamma_n^2 \|\varphi_n\|_2^2 = \\
 &= \underbrace{\|f\|_2^2}_{\text{given}} + \sum_{n=0}^N \underbrace{(c_n - \gamma_n)^2}_{\text{*****}} \|\varphi_n\|_2^2 - \underbrace{\sum_{n=0}^N c_n^2 \|\varphi_n\|_2^2}_{\text{given}}
 \end{aligned}$$

***** To obtain the min distance we must put $\gamma_n = c_n$, which completes the proof.

The Bessel's Inequality

If, in the previous slide, we replace γ_n with the Euler-Fourier coefficients c_n , then

$$\rho_2^2(\delta_N, f) = \|f\|_2^2 - \sum_{n=0}^N c_n^2 \|\varphi_n\|_2^2$$

We also know that a distance must always be non-negative

$$\rho_2^2(\delta_N, f) \geq 0$$

and in this way we obtain **the Bessel inequality**

$$\|f\|_2^2 - \sum_{n=0}^N c_n^2 \|\varphi_n\|_2^2 \geq 0$$

In the ideal situation (of no error) $\rho_2^2(\delta_N, f) = 0$, which is obtained when $\|f\|_2^2 = \sum_{n=0}^N c_n^2 \|\varphi_n\|_2^2$. In practice the error which we make by replacing a function f by the orthogonal polynomial $\sum_{n=0}^N c_n \cdot \varphi_n$ is measured by the difference

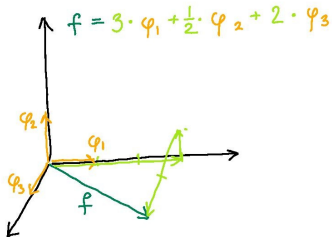
$$\|f\|_2^2 - \sum_{n=0}^N c_n^2 \|\varphi_n\|_2^2.$$

We can anticipate that the bigger N we take (ie. the degree of the polynomial is higher) the better approximation we get (consult this remark with the pictures!).



Accuracy of the Approximation - ideal situation

AGH (Parseval's condition)



$\varphi_1, \varphi_2, \varphi_3$ - form an orthonormal "basis"

PARSEVAL CONDITION

$$\|f\|_2^2 = \sum_{n=0}^{\infty} |\langle f, \varphi_n \rangle|^2$$

in this case it "looks" like:

$$\|f\|_2^2 = \underbrace{|\langle f, \varphi_1 \rangle|^2}_3 + \underbrace{|\langle f, \varphi_2 \rangle|^2}_{\frac{1}{2}} + \underbrace{|\langle f, \varphi_3 \rangle|^2}_2$$

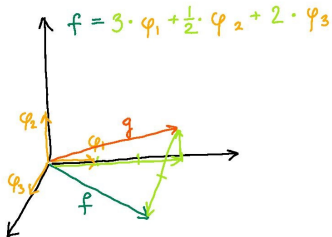
"length of the vector f"

NO "DIRECTION" IS MISSING IN THE REPRESENTATION of f
(notice that this is infinitely dimensional space, so if we miss one out of infinitely many directions still we have infinitely many directions)
What if one direction is missing? → BESSEL INEQUALITY



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Accuracy of the Approximation - realistic situation (Bessel inequality)



$\varphi_1, \varphi_2, \varphi_3$ - form an orthonormal "basis"

BESSEL INEQUALITY

$$\|f\|_2^2 - \sum_{n=0}^N | \langle f, \varphi_n \rangle |^2 \geq 0$$

in this case it "looks" like:

$$\|f\|_2^2 \geq | \langle f, \varphi_1 \rangle |^2 + | \langle f, \varphi_2 \rangle |^2$$

"length of the vector f"

$\|g\|_2^2$ (projection to φ_1, φ_2 "plane")

DIRECTION φ_3 IS MISSING IN THE REPRESENTATION OF f
If the vector f has significant component in direction φ_3 then the distance $\|f\|_2^2 - \|g\|_2^2$ is significant.

g is called THE LEAST SQUARE APPROXIMATION, it is the best approximation of f which we can get having only φ_1 and φ_2 at our disposal (eg. finite computer representation, rather than ideal, infinite one)



Condition Defining the Family of Functions AGH "representable" by the Fourier Series

Parseval's Condition

An orthonormal sequence (φ_n) of functions defined on $\langle a, b \rangle$ is called **complete** if for any square integrable functions f the **Parseval's condition** holds

$$\|f\|_2^2 = \sum_{n=0}^{+\infty} c_n^2.$$

"Completeness" of the family of functions. Assume there exists an orthonormal sequence (φ_n) among functions from some fixed family and any f from this family satisfies **Parseval's condition**. Then within this family, there is no such function which would be orthogonal to every φ_n from the orthonormal sequence (except $f = 0$ which is not interesting). Speaking informally, as in the Pythagorean thm, any function f from this family being a "hypotenuse" can be expressed as the sum of squares of mutually orthogonal "sides" ie. "base" functions $c_n \cdot \varphi_n$, but in this case there are infinitely many "sides".



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Where could we truncate Fourier series
and still get a good approximation?

Consequences of Parseval's condition

Any function that satisfies Parseval's condition can be well approximated (in the sense of the quadratic norm) by just taking an orthogonal polynomial with the Euler-Fourier coefficients.

If you feel the need for formal explanation of this fact it is enough to realise that orthonormal polynomials with Euler-Fourier coefficients are partial sums of the Fourier series, and consult the Riesz-Fischer thm for their convergence.